

ESP32 Microcontroller

Project Configuration Reference ? CBU Gyroscope Project 2026

Core Specifications

Processor	Xtensa dual-core LX6, up to 240 MHz
RAM	520 KB SRAM
Flash	External (typically 4 MB)
Operating voltage	3.3 V
Supply voltage (USB)	5 V via USB, regulated on board

Wireless Connectivity

Wi-Fi standard	802.11 b/g/n (2.4 GHz)
Bluetooth	BT 4.2 + BLE
Usage in this project	Wi-Fi for WebSocket telemetry to server

Peripheral Interfaces Used in Project

I2C	Used for MPU6050 communication (SDA / SCL)
PWM (LEDC)	Used for ESC motor speed control
UART (Serial)	Used for USB bridge to Node.js server
GPIO	Digital I/O for status LEDs and triggers

Project Configuration

Firmware framework	Arduino IDE (C++)
Sensor poll rate	50 Hz (MPU6050 via I2C)
PWM frequency to ESC	50 Hz, 1000-2000 us pulse
Telemetry protocol	JSON over serial (USB bridge)
Safety action	Emergency stop command via serial on threshold breach

Specification summary compiled for CBU Final Year Project 2026.